

JUAN CARLOS MARTINEZ FERNANDEZ

PHYSICAL ENGINEERING STUDENT

[LinkedIn](#)

Popayán-Cauca-Colombia

IA Engineer| Data Scraping| Data Engineer| Data Science| Python programmer| Data Analysis| Researcher|
Web Developer

PROFILE

Professional with experience in database management and data science projects using Python. I have worked on prominent projects using Kaggle, Physionet, GitHub, thanks to this I possess data extraction, cleaning, validation and visualization skills. I like to be an effective collaborator when working in a team, with leadership and project management skills. I have additional skills in MATLAB, Simulink, AppDesigner, Excel, Google Sheets and Overlaf. Finally, I am a very active person who likes to play sports and listen to learn new things about people or skills.

SKILLS

Technical Skills:

Databases: SQL (MySQL), Excel, CSV
Excel: (COUNT, COUNTIF, COUNTIFS, SUM, VLOOKUP, IF, Pivot Tables, Filter, Segment)
Computing (computer equipment maintenance)
Programming and Versioning: Python (intermediate), C and C++ (intermediate), Git and GitHub
Data Acquisition: Beautiful Soup, Selenium
Data Analysis: Matplotlib, Seaborn, Pandas, Numpy, Sklearn
Deep Learning: Tensor Flow, Keras, Yolo
Visualization: PowerBI
Web Development: HTML, CSS, Tailwind, Next.js, FileZilla, Hero UI, Hostinger, Google SEO, Meta Business

Soft Skills

- Teamwork
- Customer service
- Goal achievement
- Listening skills
- Planning
- First aid knowledge
- Interest in further training

EXPERIENCE

FullStack web developer [August 2025]

- Gather the requirements requested by the client for their website, delivering a mock-up of the page in Framer.
- Develop all sections of the website using HTML, React, JavaScript, and TypeScript for frontend layout. Style the page using Google fonts, shapes, colors, buttons, and animations with Tailwind CSS. Create custom images with the company logo using AI and Canva to edit them, create some of the paragraphs in the sections, and create a functional contact form configured with the company's email.
- I deployed the website using FileZilla and configured the hosting with Hostinger, giving the client more visibility in digital media.

PYTHON WEB DEVELOPER [MAY 2025].

- Take part in designing the UI and UX of a congress website.
- Develop the front-end sections. Use Python technology called Reflex, which allows you to use HTML, CSS, JavaScript, and Tailwind.

ACQUISITION OF DATA FROM NOTARY WEBSITES [APRIL 2025]

- Create a Python script for a client who needed to extract contact details from over 48,000 notaries listed in a web directory.
- Use tools such as Selenium, BS4, and Request to acquire the information, and use Pandas and Numpy to process the data.
- Everything was organized and delivered in a CSV file.

DESIGN OF A SOFTWARE TOOL USING MACHIN LEARNING TO KNOW THE HEALTH OF AN ATHLETE'S HEART THROUGH ECHOGRAM SIGNALS - [February 2024 - June 2024].

- In this project, databases such as Kaggle and Physionet were used to obtain data. The obtained files, in XML and Dat formats, were processed through Python to extract the necessary features. These features were used to train a neural network that, using previously investigated parameters, classified an athlete's heart.
- In addition, this neural network was integrated into a graphical interface developed with AppDesigner in Matlab. The interface not only allows to make predictions, but also to visualize the ECG graph and to know the different types of cardiac complexes that an athlete can experience.
- In this project, my main area of performance was the ETL (extract, transform and load) process of the data using Python.
- I provided support in the review and bug fixing of the backend, as well as in the design of the frontend.

DATA ACQUISITION FROM A DOCTORS' WEBSITE - [July 2024 - July 2024]

- I was given a link and a list of features that the client needed to acquire (Name, address, query cost, etc.) from this website. The first thing was to create a script to test the extraction using pandas, BS4, Requests and then do it massively on each page of the website. With the data obtained, a final review and cleaning of the data is done to deliver them in Excel format.

DISEASE DETECTION IN MANGO LEAVES BY IMAGE PROCESSING USING MATLAB

- The project in principle was to develop an artificial vision algorithm for the detection of different diseases in the leaves of mango crop, it was developed in MATLAB.
- My main work in this project consisted of performing the segmentation of the leaves, managing to detect the contour of the leaf and the contours that were identified as disease, this through various segmentation filters and applying basic kernels taking into account the color theory in RGB, HVS, YCbCr planes.

EDUCATION

Google Cloud Foundations

Google - 2025

Excel Mid-Avanced

Santander Open Academy - 2025

PowerBI

Santander Open Academy - 2025

Engineering Physics (last year)

Universidad del Cauca

C++ PROGRAMMING

SENA - 2021

LANGUAGES

Spanish

Native

English

middle B1